ENGLISH EDITION • COMPLETE SLIDE TEXTS (SLIDES 01–35)

Renner Masterclass & Conference 2025 • Beijing & Toyohashi

LOUIS RENNER

The Heart of the Piano: Over 140 Years of Tradition, Unified Manufacturing in Meuselwitz & Precision Hammerheads

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SPEAKER:	EVENT:	HEADQUARTERS:
Clemens Kunert	**Beijing & Toyohashi Conference 2025**	**Louis Renner GmbH • Meuselwitz, Germany**

INTRODUCTION • EXECUTIVE LEADERSHIP & PLANT MANAGEMENT

Your Contacts at Louis Renner

Personal dialogue from masters for masters — directly from our unified headquarters in Meuselwitz

EXECUTIVE MANAGEMENT

LOUIS RENNER · SINCE 1882

Erik Stieler

CEO / Managing Director

Overall corporate leadership, strategy & international piano-maker partnerships

- **Phone:** +49 3448 752889-16
- **Mobile:** +49 1522 3992381
- **E-mail:** c.kunert@louis-renner.de

> LOUIS RENNER · MADE IN GERMANY

PLANT MANAGEMENT • SPEAKER

MEUSELWITZ HEADQUARTERS

Clemens Kunert

Plant Manager / Werkleiter

Responsible for the Meuselwitz headquarters plant

- **Phone:** +49 3448 752889-12
- **Mobile:** +49 160 8062448
- **E-mail:** m.stein@louis-renner.de

> THE HEART OF THE PIANO SINCE 1882 — Louis Renner GmbH · Hainstraße 1 · 04610 Meuselwitz · Germany

Structure and Chapters at a Glance

From our Meuselwitz headquarters and Wurzen felt manufacturing to hammerhead production, the Yamaha Kit, and voicing

CHAPTER 1

Slides 4–5

History & Meuselwitz

Over 140 years of action and hammerhead tradition since 1882, plus construction insights and a drone flight through our unified Meuselwitz plant.

CHAPTER 2

Slides 6–11

Wurzen Felt Production

Filzfabrik Wurzen since 1783, premium wool blend, carding, defined fulling density, and the impact of leaving the felt sheets unsanded.

CHAPTER 3

Slides 12–20

Hammerhead Production

Felt storage, Weickert Special SEM analysis, cutting, hammer mouldings & hammer weight, pressing and gluing, CNC contour milling and precision surface calibration, plus riveting and boring.

CHAPTER 4

Slides 21–26

Yamaha Kit (C1–C3)

Ready-to-mount C1–C3 upgrade in 4–5 hours on site, time-lapse installation, pianist feedback & acoustic analysis (H_1–H_{12} to C_8).

CHAPTER 5

Slides 27–35

Voicing & Open Q&A

A_1 overtone buildup, complex spring, voicing model, crown and string plane, fine voicing, lacquer & ironing, plus Academy & open Q&A.

140+ Years of Tradition & Perfection (1882 – Today)

From a Stuttgart artisan workshop to the global benchmark for piano actions & hammerheads

ALL 9 HISTORICAL MILESTONES (1882 – TODAY)

FOUNDING

#01

1882: Founded in Stuttgart – Starting with Piano Actions

In October 1882, Louis Renner founded his workshop on Heusteigstrasse in Stuttgart as a dedicated supplier of precision piano actions.

- **Specialized Precision:** Focus on precisely fitted piano actions.
- **Historic Root:** Foundation of the manufacturing heritage united in Thuringia since 1990.

> ✦ October 1882 • Founding of the Stuttgart Action Workshop
>
> *“From the 1882 Stuttgart workshop to the 1990 acquisition of action manufacturing, Meuselwitz unites the major lines of German action making.”* — Louis Renner Chronicle

HAMMERHEADS

#02

1906: Launch of In-House Hammerhead Manufacturing

Starting in 1906, Renner produced its own hammerheads — precisely matched to the geometry and mass of its piano actions.

- **Matched System:** Hammerheads and actions engineered together under one roof for the first time.
- **Custom Presses:** In-house gluing presses ensure consistent felt hoop tension across every set.
- **Concert Benchmark:** Later adopted by Steinway & Sons and leading grand piano makers worldwide.

> ✦ Since 1906 • In-House Hammerhead Production with Custom Presses
>
> *“From Cristofori’s leather-capped roll (1700) to the tensioned felt hammer: the hammerhead directly shapes the piano’s tone.”* — Louis Renner • Hammerhead Production since 1906

GRAND ACTIONS

#03

1911: Individual Electric Drives & Grand Piano Actions

In 1911, Oscar Renner launched complete grand piano action manufacturing and converted machinery from belt shafts to hundreds of individual electric motors.

- **Érard Repetition Principle:** Double escapement as the engineering basis of modern grand actions.
- **Complete Portfolio:** Grand actions, upright actions, and hammerheads from a single source.
- **Direct Electric Drives:** Reduced machine vibration and higher repeatability via individual motors.

> ✦ 1911 • First Renner Grand Actions & Individual Electric Drives
>
> *“To unite all branches of piano action manufacturing in one hand, the production of grand piano actions was launched in 1911.”* — Oscar Renner • Company Chronicle 1911

DESTRUCTION

#04

1944: Wartime Destruction in Stuttgart & Start of Reconstruction

During the air raids of July 24/25, 1944, the Stuttgart factory buildings burned down completely; right afterward, the workforce began rebuilding.

- **July 1944:** Complete loss of the Stuttgart main plant and production machinery.
- **Immediate Restart:** Salvaging usable tooling and rebuilding workshops step by step.
- **Skilled Workforce:** Retained craftsmanship ensured the survival of precision production.

> ✦ July 24/25, 1944 • Destruction of Main Plant & Rebuilding
>
> *“Despite destruction and great obstacles: Renner will be rebuilt.”* — Wilhelm Megenhardt • Chronicle 1944

3RD GENERATION

#05

1951–1955: Dr. Manfred Renner & Reconstruction Completed

In 1951, Dr. Manfred Renner assumed 3rd-generation leadership, completed rebuilding by 1955, and expanded international exports.

- **3rd Generation (1951):** Dr. Manfred Renner takes over management of the family company.
- **Return to HQ (1955):** Full commissioning of the newly rebuilt Stuttgart main factory.
- **International Growth:** Supplying premier upright and grand piano makers worldwide.

> ✦ 1951–1955 • Dr. Manfred Renner & Rebuilt Stuttgart HQ
>
> *“By 1955, reconstruction of the Stuttgart plant was completed to its full historic scale.”* — Louis Renner Chronicle • 1955

REUNIFICATION

#06

1990: German Reunification & Acquisition of the Zeitz Action Plant

In 1990, Renner acquired Piano Union’s action plant in Zeitz — securing skilled specialists and production capacity while laying the foundation for Meuselwitz.

- **Zeitz Acquisition (1990):** Integration of the experienced Piano Union action workforce.
- **Major OEM Capacity:** Reliable supply for key customers including Schimmel (>40 pianos/day).
- **Central German Base:** Direct predecessor to today’s Meuselwitz headquarters.

> ✦ 1990 • Zeitz Acquisition & Long-Term Competitiveness
>
> *“Choosing the Zeitz site in 1990 secured our key piano-making partners and opened our future in Central Germany.”* — Louis Renner Chronicle • 1990

TWO CORE SITES

#07

2011: Consolidation into Two Core Sites – Gärtringen & Meuselwitz

In 2011, Renner consolidated its previously dispersed workshops into two modern facilities: Gärtringen and Meuselwitz.

- **Move to Meuselwitz:** Relocation from Zeitz into a modern 6,800 m² production hall.
- **Meuselwitz Plant:** Manufacturing of grand/upright actions, hammerheads, and components.
- **Gärtringen Facility:** Specialized woodworking and administration.

> ✦ 2011 • Two Modern Plants: Gärtringen & Meuselwitz (6,800 m²)
>
> *“In 2011, previously dispersed facilities were consolidated into two efficient locations: Meuselwitz and Gärtringen.”* — Louis Renner Chronicle • 2011

STEINWAY & SONS

#08

2019: Acquisition by Steinway & Sons with Full Operational Autonomy

In 2019, owner Clemens von Arnim entrusted Renner to Steinway & Sons — safeguarding jobs, quality standards, and open supply to piano makers worldwide.

- **Operational Autonomy:** Renner continues to supply all upright and grand piano makers worldwide.
- **Safeguarding Staff:** Long-term industrial stewardship chosen over higher financial bids.
- **In Grateful Memory († 2024):** Honoring Clemens von Arnim for his responsible leadership.

> ✦ 2019 • Steinway & Sons & In Memory of Clemens von Arnim († 2024)
>
> *“We share the same commitment to quality — and it is essential that Renner continues to operate independently for customers worldwide.”* — Guido Zimmermann • Steinway & Sons Europe

MEUSELWITZ HQ

#09

Today: All Manufacturing United at the Meuselwitz Headquarters

Following project launch in October 2025, action, woodworking, and hammerhead production are now fully united under one climate-controlled roof in Meuselwitz.

- **Short Paths Under One Roof:** From timber cutting to finished hammerheads and complete piano actions.
- **Modern Infrastructure:** New central dust filtration plant and year-round hall climate control.
- **Gratitude to Gärtringen:** Deep appreciation to the Gärtringen team for decades of precision work and knowledge transfer.

> ✦ Meuselwitz Headquarters • Actions & Hammerheads Under One Roof
>
> *“Experience is our foundation — reproducible precision is our commitment to every instrument.”* — Clemens Kunert • Louis Renner Meuselwitz

Construction Documentary: Building the Unified Meuselwitz Plant

Chronological photo journey (11.2025 – 09.2026) followed by an interior drone flight through production

ALL 14 MEUSELWITZ CONSTRUCTION PHASES & INTERIOR FINALE

PHASE 1: HALL PREPARATION & GUTTING
01 / 14

Project Kickoff: Site Gutting

Gutting and structural preparation of core hall lines for the future administration and reception wing.

PHASE 2: FOYER DEMOLITION & CLEARING
02 / 14

Foyer & Entrance Area Demolition

View from the foyer toward felt production: cleared large-scale areas for the new manufacturing infrastructure.

PHASE 3: UTILITIES & GROUNDWORK
03 / 14

Utilities & Fresh Water Supply

Precision trenching for fresh and industrial water lines – conducted seamlessly without disrupting ongoing production.

PHASE 4: ACTION ASSEMBLY LINE SETUP
04 / 14

Upright Action Assembly Installation

Former warehouse areas are transformed into a state-of-the-art production line for upright piano wippens.

PHASE 5: MODERN WAREHOUSE LOGISTICS
05 / 14

Smart Storage Logistics & Vertical Lifts

Maximum space efficiency: installation of automated vertical shuttle towers storing tens of thousands of precision components.

PHASE 6: FILTRATION SYSTEM FOUNDATION
06 / 14

Reinforced Foundation for Filter System

Reinforcement and concrete pour for the heavy-duty foundation of the 40-ton, 12-meter-tall central filtration plant.

PHASE 7: CRANE LIFT OF FILTER MODULES
07 / 14

Delivery & Crane Lift of Filter Modules

Heavy-duty logistics with a 70-ton mobile crane: millimeter-accurate positioning of multi-ton filtration components.

PHASE 8: CRANE ASSEMBLY OF FILTER TOWER
08 / 14

Crane Assembly of 12 m Filter Tower

Assembly of the 12-meter-tall upper filter tower section onto the vibration-isolated steel frame.

PHASE 9: HAMMERHEAD RELOCATION
09 / 14

Relocation of Hammerhead Production

Relocation of hammerhead production into the newly built, climate-controlled hall section in Meuselwitz.

PHASE 10: INDUSTRIAL FILTER ASSEMBLY
10 / 14

Assembly of 120,000 m³/h Filter Plant

With a 120,000 m³/h airflow capacity, the new filter plant ensures dust-free, dimensionally stable production.

PHASE 11: DUCTWORK & OVERHEAD TRUSSES
11 / 14

Extraction Trusses & Hall Ductwork

Installation of large-diameter overhead ductwork for continuous dust extraction across all woodworking machines.

PHASE 12: ROOFTOP HVAC & CLIMATE PLANT
12 / 14

Rooftop Plant: Process Air & HVAC

Year-round constant temperature and 45–50% relative humidity for dimensionally stable wood and felt processing.

PHASE 13: HUMIDITY CONTROL & CONDITIONING 13 / 14

Precision Humidity Control for Wood & Felt

Controlled humidification prevents checking in tonewoods and dimensional drift in hammer felt.

PHASE 14: MOLDING SHOP & FULL OPERATION 14 / 14

Molding Line Setup & Full Operation

Final department relocated: all wood, action, and hammerhead manufacturing now operates under one roof in Meuselwitz.

INTERIOR DRONE FLIGHT • MEUSELWITZ PLANT (2026) 2026

The Completed Meuselwitz Plant – Interior Drone Flight

Filzfabrik Wurzen (FFW) – The Birthplace of Piano Hammer Felt

Wurzen 1783 • From the first piano felt in 1847 to the Weickert Spezial partnership with Renner

ALL 6 HISTORICAL FFW WURZEN ERAS (1783 – TODAY)

ORIGIN WURZEN 1783

`Wurzen 1783`

Founding of the Felt Manufactory in Wurzen

The world's oldest industrial felt manufactory – rooted in Saxony's Mulde valley

- **Founding in 1783:** Johann Dankegott Weickert establishes the Saxon wool and felt tradition in Wurzen.
- **Fine Felt Pioneer:** The artisan shop grows into a leading center for premium technical virgin wool felts.
- **Piano Focus:** Early dedication to highly resilient felt for the emerging pianoforte industry.

> 🏛 World's oldest industrial felt manufactory in Wurzen

INVENTION 1847

`1st Piano Felt`

Germany's First Viable Piano Hammer Felt

Patzschke & Weickert revolutionize European piano craftsmanship in 1847

- **Felt Replaces Leather:** In 1847, uneven leather hammer coverings are successfully replaced by homogeneous felt.
- **Strong Partnership:** Weickert's enterprise and Patzschke's felt expertise establish international renown.
- **Soft River Water:** The naturally soft, low-mineral Mulde water ensures pristine, resilient wool fibers.

> ✦ Milestone: Wurzen felt permanently replaces leather in pianos

LONDON & PARIS 1851

`World Expo`

World Expos & Gold Medals in London & Paris

International gold medals at the World Exhibitions in Leipzig, London, and Paris

- **World Expo Gold:** Top honors in Leipzig (1850), London (1851), and Paris (1867) establish global acclaim.
- **Royal Visits:** Saxon monarchs personally inspect and praise the superlative craftsmanship of the Wurzen mills.
- **Europe's Top Choice:** Premier piano builders across Vienna, Paris, and London adopt Weickert felt exclusively.

> 🏅 Gold Medals London 1851, Paris 1867 & Leipzig 1850

EXPANSION 1861

`Wurzen Mill`

Industrial Scale at Mühlgraben in Wurzen

Steam power, modern fulling mills, and the clean waters of the Mulde river

- **Industrial Scale:** In 1861, modern facilities with proprietary steam power are erected at Wurzen's Mühlgraben.
- **Precision Fulling Mills:** Steam-powered machinery guarantees unprecedented density and batch uniformity.
- **Gentle Wool Scouring:** Mineral-poor river water preserves the natural cuticle scale structure of the Merino wool.

> ⚙ Steam fulling & natural river water at Wurzen Mühlgraben

WORLD LEADER 1906

`World Leader`

Global Market Leader for 200,000 Pianos Annually

Director Patzschke, his master craftsmen & 350 specialists at Wurzen (1906)

- **Market Leader by 1906:** Wurzen equips up to 200,000 pianos and grands worldwide with hammer felt each year.
- **Global Benchmark:** Steinway & Sons New York and Europe's premier manufacturers rely on Wurzen felt.
- **Masters & Director Patzschke:** Historic 1906 photograph standing before the monumental tapered felt block.

> 🌎 Global reference standard for Steinway & Sons New York

TODAY • RENNER PARTNER

`Weickert Special`

Renaissance of “Weickert Spezial” & Partnership with Renner

Historic Wurzen mill & direct geographical proximity to Renner's factory in Meuselwitz

- **Revived in 2010:** The Brand family restores the historic 1903 Wurzen formula for concert-grade hammer felt.
- **Geographical Proximity (~65 km):** Wurzen and Renner's Meuselwitz plant share Central Germany's piano craft cluster.
- **Direct Master Collaboration:** Short travel times enable face-to-face calibration of felt density and elasticity on-site.

> 📍 Only 65 km: Direct partnership Wurzen ↔ Renner Meuselwitz

100% PURE MERINO VIRGIN WOOL

`Micro-Texture: Fulling Compression`

The Secret of Weickert Felt

Purely mechanical fulling without chemical additives – for rapid rebound and clear concert tone

- **Exclusive Wool Blend:** Selected long and crimped wools from Argentina, South Africa, and New Zealand form an ultra-tensile fiber matrix.
- **Pure Fulling without Resins:** Densification solely through Mulde river water, heat, and pressure – zero dampening resins or glues.
- **Permanent Elastic Springback:** Natural keratin springs rebound in milliseconds – liberating the entire overtone harmonic spectrum.
- **Proximity to Meuselwitz (65 km):** Only ~65 km distance ensures direct exchange between felt and hammer masters and immediate availability.

Pure Merino Wool & the Secret of Fiber Blending

Uncompressed overseas raw wool and tailored fiber recipes as the foundation of hammerhead felt

RAW WOOL SELECTION

Stage 01

Uncompressed Overseas Merino Wool & Intact Fiber Structure

- **Uncompressed Bale Delivery:** Loosely packed shipping preserves the microscopic keratin scale structure intact.
- **Pronounced Fiber Crimp:** Millions of natural micro-springs ensure optimal interlocking during fulling and rapid rebound.
- **Natural Lanolin:** Pure wool grease and gentle scouring keep the keratin fibers supple for decades.

> **UNCOMPRESSED RAW WOOL:** Loosely delivered Merino wool retaining its strong natural fiber crimp

FIBER FORMULA

Stage 02

Gentle Fiber Opening & Calibrated Blend Formula

- **Calibrated Fiber Blend:** Precise balance of long tensile fibers (for the crown) and fine filler wool (for core density).
- **Multi-Stage Opening:** Gentle plucking and pneumatic aeration separate fiber tufts cleanly into individual fibers.
- **Climate Conditioning:** Controlled moisture uptake prevents static charge, fiber breakage, and density spots.

> **FFW BLENDING CHAMBER:** Pneumatic loosening and blending of wool batches at FFW Wurzen

Carding: Parallel Fiber Alignment into Fine Fleece Webs

Defined fiber orientation as the basis for uniform elasticity and predictable needle voicing

CARDING & FLEECE

`FIBER ORIENTATION`

Drum cards comb fibers in parallel alignment

Deliberate longitudinal and transversal alignment for predictable strike elasticity

- **Carding:** Fine wire rollers gently comb the opened wool, untangling and aligning the fibers in parallel.
- **Defined Fiber Course:** The gossamer-thin fleece web receives a precisely controlled primary fiber direction.
- **Controlled Layering:** Superimposed fleece webs build target sheet thickness, tensile strength, and wedge taper.
- **Foundation for Voicing:** Only ordered fibers part cleanly and predictably when voiced with needles later on.

> ✦ Parallel fiber guidance is the cornerstone of precise voicing

FLEECE STRUCTURE

`Fleece Web: Ordered Fibers`

Prerequisite for Voicing

Only parallel aligned fibers respond predictably and uniformly to needle penetration during voicing

- **Gliding Needling:** Voicing needles part parallel fibers rather than severing them.
- **Fluid Density Gradients:** Cross-layered stacking prevents delamination under extreme touch forces.
- **Maximum Fiber Length:** Gentle carding prevents fiber breakage and ensures maximum tensile strength of the fleece.

> Carding Merino wool into uniform fleece webs with defined fiber alignment

The 4 Pillars of Fulling: Steam, Pressure, Motion & Experience

Purely mechanical-thermal fiber interlocking with soft Mulde water – 100% free from glues and resins

HAMMER MILL

Pure mechanical felting without extraneous additives

Water, heat, pressure, and rhythm permanently interlock fibers

- **1. Live Steam:** Opens microscopic cuticle scales of fibers and renders keratin supple.
- **2. Mechanical Pressure:** Continuously compresses fleece layers, initiating progressive density increase.
- **3. Oscillating Motion:** Frictional rubbing interlocks natural barbed scales of fibers inextricably.
- **4. Artisan Know-How:** Sensory feel for the exact moment of completion governs elasticity and acoustic resonance.

> ✦ Solely soft Muldental water and heat – 100% chemical-free

FULLING COMPRESSION

`Micro-Texture: Fulling Compression`

Physical Compaction

Scales of wool hairs interlock micromechanically like burrs

- **No Damping by Glues:** Pure mechanical felting preserves free oscillation of wool fibers.
- **Permanent Elasticity:** Natural keratin springs rebound within milliseconds without fatigue.
- **Maximum Overtone Purity:** Soft Muldental water prevents mineral deposits inside fiber pores.

> Fulling layered fleece webs under steam, pressure, and motion into a homogeneous felt sheet

HOT STEAM

1. Live Steam

Opens the microscopic cuticle scales of the wool fibers and renders the keratin supple.

MECHANICAL PRESSURE

2. Mechanical Pressure

Compresses the layered fleece webs uniformly to initiate progressive densification.

OSCILLATION

3. Oscillating Motion

Rhythmic friction permanently interlocks the natural barbed scales of the fibers.

CRAFTSMANSHIP

4. Master Experience

Judging the exact stopping point governs natural springiness and overtone richness.

56% Fulling Density (0.56 g/cm³): The Natural Limit of Mechanical Fulling

Maximum natural springback without fiber damage or chemical over-hardening

PHYSICAL LIMIT

56% • 0.56 g/cm³

Maximum Natural Fulling Density of 56%

Felt sheets are compacted precisely to the point where fibers contact optimally

During fulling, felt sheets are compacted until the wool fibers cannot be pushed closer purely mechanically—reaching a maximum fulling density of 56% (0.56 g/cm³).

- **Maximum Fiber Compression:** Fibers cannot be mechanically compacted further without structural damage.
- **Peak Natural Spring Rate:** Corresponds to the physical optimum of inherent elasticity for concert grand touch.
- **Free from Synthetic Resins:** Eliminating chemical hardeners ensures permanent organic tone blooming.

> ✦ Max. 56% Fulling Density (0.56 g/cm³): Peak elasticity with zero hardeners

ACOUSTIC PERFECTION

FFW WURZEN

Elastic Springback Instead of Brittle Hardness

56% natural fulling density delivers brief string contact times and nuanced dynamic control.

- **Rapid String Rebound:** The hammer rebounds instantly off strings without muting tone.
- **Color-Rich Tone Spectrum:** Full blooming of fundamentals and overtones without treble harshness.
- **Stable Hammer Action:** Consistent density sustained over tens of thousands of keystrokes.

> Fulled tapered piano hammer felt sheets at Filzfabrik Wurzen • Post-fulling finishing of the mechanically fulled felt sheet in Wurzen

Final Compaction & Tapered 36-Layer Anatomy

Dimensional calibration without sanding – 2×6 continuous outer webs preserve full bending tension

SHAPING WITHOUT GRINDING

HAMMER & STONE

Manual compaction and leveling

Targeted hammer blows bring felt sheets to exact size while relaxing the fiber lattice.

- **Mechanical Hammering:** Felt hammers and polished natural stones consolidate sheets to exact dimensions.
- **Zero Sanding Removal:** The Weickert process eliminates sanding completely—not a single tension fiber is cut.
- **Unbroken Bending Radius:** When pressed around the hammer moulding, the outer skin stays smooth and tear-free.

> ✦ Intact surface fiber skin ensures maximum longevity — Fulled felt sheets with natural indentation firmness of 56 to 68 Shore-A

CONICAL FLEECE STRUCTURE & BACKBONE

Conical Taper & Tensile Stability: The Fleece Web Backbone

From thin treble on the left to thick bass on the right, shortened core webs form the wedge—enclosed by 2×6 outer webs.

- **Treble → Bass:** Shortened core webs build the thick bass on the right while keeping the treble thin on the left.
- **2×6 Outer Webs:** Continuously unsanded—absorbs all bending and tensile forces.
- **Sanding Effect:** Severed outer webs slip apart—loss of structural pre-tension.
- **✓ Unsanded • Shape-Stable:** Unsanded: 2×6 outer webs enclose the shortened core layers • Full tension retention
- **✕ Sanded • Deformation:** Sanded: Outer fleece webs severed • Fibers slip apart under bending tension

The Hammerhead Department at the Meuselwitz Plant

Organic drone tour from the climate-controlled felt warehouse across the press line to CNC finishing

CHAPTER 3 • DRONE TOUR

`4K Drone Flight`

The Hammerhead Department at the Meuselwitz Plant

Organic drone tour from the climate-controlled felt warehouse across the press line to CNC finishing

> Seamless drone flight through the hammerhead manufacturing department in Meuselwitz

Climate-Controlled Felt Storage & Assortment Diversity

Over 30 tapered top-felt formulas plus 6 standard underfelt colors and custom dyeings in stock

TOP-FELT INVENTORY

30+ Grades

Tapered FFW & Weickert Sheets in Controlled Seasoning

- **Mill-Fulled Taper:** Thick in the bass, progressively slimmer and denser toward the treble.
- **Broad Assortment:** Over 30 calibrated FFW and Weickert formulas for concert, OEM, and historic instruments.
- **Climate & Resting Period:** Regulated humidity, pheromone monitoring, and seasoning relieve fulling stresses.

> Drone flight through the climate-controlled felt sheet storage in Meuselwitz • Underfelt stock in Meuselwitz – custom colors available on request

6 STANDARD + CUSTOM COLORS

Underfelt Color & Specification Assortment

- **Red & Wine Red (CLASSIC):** Traditional Renner signature colors for European concert and prestige grands.
- **Orange (HERITAGE & CONCERT):** Classic specification for concert instruments and period-authentic restorations.
- **Yellow (MANUFACTORIES):** Heritage underfelt color of renowned Saxon and Viennese piano makers.
- **Blue, Black & Custom (MARQUES & CUSTOM):** Classic blue (e.g., C. Bechstein), sleek black, plus bespoke customer dyeings.

“Weickert Special” – Exclusive to Renner

56% maximum fulling density and elevated residual wool grease per the original 1903 Wurzen formula

FFW WURZEN × STFI LAB REPORT

56% Fulling Density

Highest Fulling Density & Protected Keratin

56% Max. Fulling Density (0.56 g/cm³ chemical-free) | 0.36% Residual Grease (ISO 3074) (+38% vs. 0.26% sample) | –65% Climate-Cycle Swelling (4.81% vs. 13.69%)

Gentle wool scouring and intensive fulling unite 56% peak density with elevated wool grease retention (STFI Report P2024 0903).

- **56% Fulling Density:** Highest purely mechanical density (0.56 g/cm³) for strong internal tension without chemical hardeners.
- **0.36% Residual Grease (ISO 3074):** Gentle scouring retains +38% more natural wool wax (vs. 0.26%) as moisture protection.
- **Verified Climate Stability:** In the 48h STFI climate-cycling test, thickness swelling drops by 65% (only 4.81% vs. 13.69%).
- **Intact Cuticle & Voicing Stability:** Epicuticle scales visible under SEM ensure strong fiber traction and lasting tonal consistency.

> ✦ Verified: Accredited STFI Test Report P2024 0903 (M. Kermer, ISO 3074 & Climate Cycle)

STFI • SEM

STFI Scanning Electron Microscope (SEM) Zoom Levels

- **1× Overview (1 mm):** SEM Overview (STFI Mik59_24 • 1 mm): Ultra-dense 3D fiber interlock with 56% maximum fulling density.
- **10× Fiber Detail (100 µm):** SEM Detail 10× (STFI Mik59_24 • 100 µm): Merino fibers with intact cuticle scales and +38% higher residual wool grease (0.36%).
- **22× Cuticle Zoom:** SEM Deep Zoom (>18×): Overlapping epicuticle scales and lanolin-protected keratin ensure fiber traction & climate stability.
- **Felt Strip W 1512 K:** Macro View: Cut and batch-coded Weickert Special top-felt strips prior to hammerhead pressing.

> Wool Fiber Cuticle Scales

Felt Preparation: Top-Felt Cutting & Underfelt Profiling

Robotic strip cutting to 1 mm tolerance and precision profile milling prior to pressing

01 • 6-AXIS ROBOTICS

`≤ 1 mm Tolerance`

Robotic Top-Felt Cutting

- **1 mm Strip Cutting:** 6-axis robot divides seasoned tapered felt sheets distortion-free across the full width.
- **Crush-Free Cold Cut:** Sharp blades part the Merino fibers cleanly without thermal glazing or edge compression.
- **Inline Batch Coding:** Sheet and order IDs marked directly on every strip for complete traceability.

> 6-axis robot performing distortion-free top-felt strip cutting

02 • PROFILE MILLING

`± 0.1 mm Tolerance`

Underfelt Profile Milling

- **Elastic Coupling Bridge:** Contour-milled underfelt transfers strike energy smoothly from the rigid hammer moulding into the top felt.
- **Feathered Tails (± 0.1 mm):** Ends milled to a ± 0.1 mm tolerance ensure a seamlessly closed hammer contour right into the high treble.
- **Balanced Press Load:** The calibrated wedge contour distributes jaw pressure evenly across both hammer flanks.

> Contour-milling the underfelt strip before loading into the glue press

The Hammer Moulding: Tonewoods, Properties & Profiles

From heavy hardwoods (hornbeam, maple) and oil-rich Kotibé to lightweight, fine-grained American walnut

HARDWOOD • IDEAL `Heavy • Stable`

Hornbeam (Weißbuche)

Very demanding to season—yet once properly dried, an ideal and uniform moulding wood.

HARDWOOD • ELASTIC `Heavy • Selected`

Maple (Ahorn)

Heavy, often more elastic hardwood—though irregular sugar deposits carry a higher risk of variance.

PREMIUM • TREBLE `Very Light • Fine-Grained`

American Walnut

Highest-grade wood: tight growth rings, long-fibered, and very light—ideal for a brilliant treble.

SPECIALTY • OIL-RICH `Heavy • Specialty`

Kotibé (Mahogany)

Comparatively heavy due to its high natural oil content—suited only for specific applications.

FORM B · US · CSB

Symmetrical Profiles

Steinway & Sons, Bösendorfer, C. Bechstein, Yamaha, Kawai, Schimmel, Blüthner, Förster, and Sauter.

> Renner hammer moulding cross-sections: Form A, Form B, Form US, and Form CSB

FORM A

Asymmetrical Profiles

Form A with offset moulding shoulder for vintage Bechstein grands and slender treble scale designs.

> Selected hammer mouldings – knot-free, straight-grained & carefully seasoned

Optimal Hammerhead Weight Relative to String Weight

From ~15 g in the bass across the −1 g break step to 5 g in the treble: balancing contact time and string deflection

CORE EQUILIBRIUM

`String Weight ⇔ Hammer Mass`

Crown Equilibrium: Contact Time τc ↔ String Deflection

When **string weight** and **hammerhead weight** match, **contact time τc** and **string deflection** reach ideal equilibrium.

- **Copper-Wound (Bass · Keys 1–23)** | **Plain Wire: 3 Strings per Note (Trichord · Keys 24–88)**
- ▲ **Simulated Too Light:** Hammerhead rebounds too early – low string deflection, thin tone.
- ▲ **Simulated Too Heavy:** Excess contact time on the string – overtones are damped, dull tone.

ALL 4 REGISTER MASS-SCALING ZONES (KEYS 1–88)

✦ MASS MATCHING ACROSS ALL 88 KEYS

From ~15 g in the Bass Across the 1 g Break Step to 5 g in Treble

Hammerhead weight must balance actual string weight: bass hammers start around 15 g and are held high as long as possible (15 → 13 g), step down by 1 g onto plain steel in the middle, and finish at 5 g in the treble.

> **Renner Engineering:** Targeted moulding tonewoods (hornbeam to lightweight American walnut), conical Weickert felt taper, precision moulding profiling, and contoured tail shaping.

① DEEP BASS & BASS SECTION (~15.0 → 13.0 G)

Heavy Copper Winding — Keeping Bass Hammer Weight High

Copper-wound bass strings weigh 90 to over 200 g—so in the bass, the hammerhead (~15 → 13 g) is always comparatively light relative to the string. Bass hammer weight is therefore held high as long as possible.

> **Renner Solution:** Substantial bass profile (15.0 → 13.0 g) with thick, tension-preserved Wurzen felt and targeted tail relief so contact time stays controlled despite high mass.

② COPPER → PLAIN STEEL TRANSITION (−1.0 G STEP)

1 g Hammer Weight Step onto Plain Steel Middle Strings

Entering the 3-string plain steel section (3 wires per note, ~31 g combined vs. 90 g in bass), total choir mass drops sharply. Because bass hammers were held high (13 g), a deliberate −1 g step to 12 g restores balance.

> **Renner Solution:** Calibrated −1.0 g weight step (13.0 → 12.0 g) right at the transition, paired with matched underfelt and crown elasticity for an even tonal bridge.

③ LAST 2 OCTAVES IN TREBLE (DOWN TO 5.0 G)

Light 3-String Choir (< 3 g) Down to the 5 g Treble Hammer

Even across all three strings per note, high-treble choir mass drops from 2.7 g (C_6) to ~0.6 g (C_8; ~0.2 g per wire) with ½T < 0.5 ms. Excess hammer mass prolongs contact and immediately self-damps the trichord.

> **Renner Solution:** Smooth taper down to 5.0 g on the final treble hammer (C_8) with lightweight, long-fibered moulding wood (e.g., American walnut) and a firm, pointed crown for instant rebound.

Pressing & Gluing to 0.1 mm

Building outer tensile tautness and inner cushion support in the glue press

GLUE PRESSING

`0.1 mm Tolerance`

Pressing in the Kettle and Heated Side Jaws

Precision glue presses unite top felt, underfelt, and hammer moulding with repeatable accuracy.

- **0.1 mm Pressing Tolerance:** Precision-guided jaws maintain exact dimensional accuracy across the entire set.
- **Kettle & Side Jaws:** The felt is pressed downward into the kettle; the side jaws then close in and press the felt firmly onto the glue surface.
- **Defined Stress Zones:** High outer tensile force meets a compressed, resilient inner cushion.

> ✦ This internal tension gradient underpins dynamic range from pp to ff • Pressing into the kettle and closing the side jaws in Meuselwitz

CNC Milling & Robotic Calibration

0.1 mm moulding milling, post-gluing trim milling, and robotic calibration of every individual hammerhead

CNC MILLING & ROBOTICS

`0.1 mm Tolerance`

Two-Stage CNC Milling & Single-Head Calibration

CNC mills first shape the hammer mouldings and trim the felt overhang after gluing—before a robotic cell calibrates each individual head.

- **1. Moulding Milling (0.1 mm):** CNC milling machines first machine the hammer mouldings to a 0.1 mm tolerance.
- **2. Post-Gluing Trim Milling:** After pressing and gluing, the felt overhang is removed and cleanly finish-milled.
- **3. Robotic Calibration:** Subsequently, a robotic cell precisely calibrates each individual hammerhead.

> ✦ 0.1 mm CNC accuracy combined with repeatable robotic single-head calibration • Drone flight across the SCM CNC milling centers and robotic calibration cell in Meuselwitz

Automated Riveting & Boring

Action-matched mounting geometry and permanent felt-tail retention

FINAL FINISHING

`ø 5.4 mm · Riveting`

Mounting Geometry for the Hammer Shank

Every individual head receives its model-specific bore geometry and mechanical lock.

- **Precision Boring Angle:** Exact depth, diameter (e.g., ø 5.4 mm), and individual boring angle per note.
- **Brass Riveting:** Permanently locks the bonded felt tails against opening under heavy concert playing.
- **Radiused Hammer Tail:** Cleanly contoured tail radii ensure reliable checking against the backcheck felt.

> ✦ Ready for accurate gluing onto the hammer shanks • Automated shank boring and brass rivet insertion

Louis Renner Yamaha Kit (C1–C3) – Seamless Drop-In

Tailored for Yamaha C1–C3: Dynamic expressiveness, brilliant overtones & deep fundamental bass.

ALL 5 RENNER YAMAHA KIT (C1–C3) MODULES

MEASUREMENT-BASED PRODUCTION · 5 YEARS R&D

`Measurement System`

Proprietary Measurement & Production System

Supreme voicing culture for new and existing Yamaha C1–C3 grand pianos.

- **Calibrated System:** 5 years of R&D for perfectly tuned felt density, elasticity, and tension.
- **Defined Target Tone:** Every set manufactured to millimeter precision for the C1–C3 acoustic optimum.
- **Tone Without Voicing:** Instant full, warm concert sound – zero manual pre-voicing needed.

> ✦ Reliable Concert Tone: Instant master result straight from the factory

OPTIMIZED INERTIA

`Weight & Touch`

Balanced Weight & Fast Repetition

Calibrated walnut moulding with fine-grained shanks for noticeably lighter key response.

- **Calibrated Walnut Moulding:** Lightweight tonewood moulding reduces inertia for agile, effortless playing.
- **Selected Hornbeam:** Fine-grained quarter-sawn wood guarantees maximum stiffness and pure vibration.
- **Pre-Tensioned Knuckles:** Glued under high tension for superior resilience and instant repetition.

> ✦ Noticeably Lighter Touch: Reduced inertia for precise dynamic control and rapid repetition

SCREW IN · ALIGN · DONE

`Screw In & Align`

Simply screw in, align – done.

Factory-bonded onto flanges at the exact angle using traditional natural hide glue.

- **1-Screw Mounting:** Custom flanges screw note by note directly into the original Yamaha rail.
- **Pre-Bored Accuracy:** Pre-drilled at the exact angle – zero measuring or boring in the workshop.
- **Glass-Hard Hide Glue:** Traditional hot hide glue cures rock-hard for lossless vibration transfer into the shank.

> ✦ Peak Process Reliability: Zero risk of drilling or gluing errors

HARMONIC-RICH CONCERT TONE

`Concert Tone`

Rich Harmonic Spectrum & Color

From warm European pianissimo to powerful projection in fortissimo.

- **Seamless Felt Strip:** Pressed from a single strip — guarantees seamless transitions across all 88 keys.
- **Articulated Bass:** Balanced mass and resilient felt for deep fundamental tone and clean note separation.
- **Bell-Clear Treble:** Lightweight moulding and targeted crown compression for a singing, carrying treble tone.

> ✦ Acoustic Elevation: Transforms harsh attack into warm, singing concert tone

HOURS INSTEAD OF DAYS

`Hours Instead of Days`

4–5 Hours on Site Instead of 3–7 Workshop Days

Traditional multi-day workshop procedure versus same-day installation and regulation on site.

- **Traditional (3–7 Days):** Transport, ungluing, boring chart, hot glue, aligning, sanding & pre-voicing.
- **Renner Kit (4–5 h):** Bolt on directly, align string level and crown – finish voicing if desired.
- **On-Site & Room Acoustics:** Zero transport risk — more time to tailor the sound to room acoustics and customer wishes.

> ✦ Peak Economic Efficiency: Work reduced from days to a few hours on site

RENNER YAMAHA KIT

`Macro: Head on Shank with Flange`

Heads & Shanks in Detail

100% drop-in pre-assembled for Yamaha C1, C2, and C3.

- **Calibrated System:** Proprietary measurement process for finished sound straight from the factory.
- **Walnut & Hornbeam:** Minimized moment of inertia for lightning-fast repetition and agile response.
- **Drop-In & Hide Glue:** Rock-hard acoustic coupling, factory pre-glued – 0 h drilling labor.
- **1-Screw Installation:** Plug & play into original rail – installation & regulation in 4–5 hours on site.

Time-Lapse Installation: Complete Conversion in One Day

Documentation of the same-day installation (10:00 AM – 6:00 PM) on site in India (10/2026)

TIMELAPSE · 10–18 H

1 Workday

Complete Action Upgrade Directly On Site

Time-lapse from India (10:00–18:00): Complete on-site conversion including removal, mounting, regulation, tuning & test play.

- **Same-Day Conversion:** Complete upgrade to the pre-assembled Renner Kit from 10:00 to 18:00 on site.
- **Plug & Play Fit:** Direct mounting of pre-calibrated flanges and shanks onto the original rail.
- **On-Site Regulation:** Precise alignment, blow distance, let-off, and checking set right at the grand.
- **Zero Transport Risk:** The grand stays in place and is voiced directly to the room's acoustics.

▸ 10:00–18:00: From removal to a concert-ready grand on site

Pianist Feedback: Pianist Raaga in Action

Immediate artist evaluation of tone color, dynamic range, and repetition on the converted Yamaha grand

ARTIST SOUND PROOF

100% Concert

Immediate Artist Verdict After Installation

Pianist Raaga tests the upgraded Yamaha grand first-hand for tonal warmth, projection, and touch.

- **Singing Concert Tone:** Audible shift from a hard attack to a warm, rich, projecting sound.
- **Blooming Treble:** Bell-like melody projection paired with clean chord clarity in the bass.
- **Nuanced Dynamics:** Effortless shading and tonal control from whisper-soft pp to full ff.
- **Effortless Repetition:** Crisper key response and rapid return thanks to calibrated hammer mass.

> ✦ Immediately audible shift to European warmth and touch control

Sound Spectrum Comparison: Original vs. Renner Yamaha Kit

Original Hammerheads (0:00–5:17) vs. Louis Renner Yamaha Kit (5:18–11:42)

5 SPECTRAL FREQUENCY BANDS (ORIGINAL VS. RENNER YAMAHA KIT)

1. BASS (50–250 HZ)

`-3.28 dB (-53% Mud)`

Eliminates low-frequency boominess from worn hammers for crystal-clear bass chord separation.

2. BODY (250–800 HZ)

`-0.07 dB (100% Stable)`

Warm fundamental body remains completely intact — higher brilliance without losing fullness.

3. PRESENCE (800–2,500 HZ)

`+2.53 dB (+79% Energy)`

Primary gain in the singing register (peak +3.4 dB at 1.45 kHz) for a projecting concert tone.

4. BRILLIANCE (2.5–6.0 KHZ)

`+1.87 dB (+54% Energy)`

Bell-like overtones and richer timbres via Weickert Special felt and flatter spectral slope.

5. AIR (6.0–12.0 KHZ)

`+0.31 dB (+7% Energy)`

Crisp attack definition without metallic harshness in the extreme treble.

2-OCTAVE METRICS

Metrics in 2-Octave Window

- **2-Octave Window:** A_0–C_8 (25 Hz – 10 kHz)
- **Overtone Gain (H2–H8):** +2.53 dB (0.8–2.5 kHz) / +1.87 dB (2.5–6.0 kHz)
- **Fundamental H1 (f_0):** 100% (−0.07 dB, 250–800 Hz)
- **Overtone Roll-Off:** −6.45 dB/kHz vs. −7.38 dB/kHz (−12.6%)

Partial Distribution (H_1–H_{12}) & 24-Semitone Comparison

Overtone architecture per 2-octave window: More warmth and brilliance

PARTIAL SERIES H_1–H_{12}

Harmonic Partial Distribution (up to 6 kHz)

- **Roll-Off (H_1→H_8):** −6.45 dB/kHz vs. −7.38 dB/kHz
- **Even (H_2, H_4, H_6, H_8):** Warmth & Body
- **Odd (H_3, H_5, H_7):** Brilliance & Core
- **H_1–H_{12}:** H1 • Fundamental ($1 \cdot f_0$) • H2 • Octave ($2 \cdot f_0$) • H3 • Twelfth ($3 \cdot f_0$) • H4 • 2nd Octave ($4 \cdot f_0$) • H5 • Major Third ($5 \cdot f_0$) • H6 • Fifth ($6 \cdot f_0$) • H7 • 7th Partial ($7 \cdot f_0$) • H8 • 3rd Octave ($8 \cdot f_0$) • H9 • Ninth ($9 \cdot f_0$) • H10 • Tenth ($10 \cdot f_0$) • H11 • 11th Partial ($11 \cdot f_0$) • H12 • 12th Partial ($12 \cdot f_0$)

24 CHROMATIC SEMITONES

Note-by-Note Overtone Energy (H_2–H_8)

- **Register:** A0–A2 (Bass) | A1–A3 (Tenor) | A2–A4 (Middle) | A3–A5 (Treble 1) | A4–A6 (Treble 2) | A5–A7 (Brilliance) | Full (A0–C8)
- **Mode:** Relative H_1 | Δ Gain | Absolute (dB)

4 Tonal Gains & Overtone Retention up to C8

Measurable gain in the singing and brilliance registers plus flatter overtone roll-off up to 4,300 Hz

01 • CANTABILE

Singing Register (0.8–2.5 kHz)

Up to +3.4 dB at 1.45 kHz for a warm, singing concert tone.

02 • PROJECTION

Treble Brilliance (2.5–6.0 kHz)

Carrying overtone brilliance without harshness thanks to Wurzen Weickert felt.

03 • CLARITY

Bass Precision (20–250 Hz)

Reduced low-end boominess for clearly separated bass chords and defined fundamentals.

04 • OVERTONE RETENTION

12.6% Flatter Roll-Off & Crisper Attack

Stronger upper partials (−6.45 vs. −7.38 dB/kHz) and +0.42 dB attack crest.

OVERTONE ROLL-OFF (TO C_8 = 4,300 HZ)

Overtone Roll-Off by Touch (pp–ff) & Frequency Calculator

- **Original (−7.38 dB/kHz)** vs. **Yamaha Kit (−6.45 dB/kHz)**
- **✦ Pitch Gain Calculator (up to C_8 = 4,300 Hz):** D#$_7$ (2,500 Hz): **+2.33 dB** Net Gain | C_8 (4,300 Hz): **+4.00 dB** Net Gain

Partial Structure on A_1 (55 Hz): Felt Hardness & Strike Point

How felt elasticity and strike ratio (x_0 / L) shape partials H_1–H_{12} in decibels (dB)

FELT HARDNESS PROFILES & YOUNG-HELMHOLTZ STRIKE-POINT LAW (A_1 = 55 HZ)

☁ SOFT

Soft Felt: Longer τc Tames Upper Partials

Deeper cushioning (**τc ≈ 2.85 ms**) lowers H_6–H_{12} by **−4.6 dB** — H_1 and H_2 bloom warmly.

✦ CONCERT IDEAL

Concert Balance: Rich Core & Controlled Crown

Balanced elasticity (**τc ≈ 1.95 ms**) blends full bass foundation (H_1–H_4) with singing H_5 and H_6.

✦ HARD

Hard Felt: Short τc Boosts Upper Partials

Fast rebound (**τc ≈ 1.25 ms**) lifts H_6–H_{12} by **+4.8 dB** (H_{12}: **+6.4 dB**) — brighter, shorter sustain.

STRIKE POINT X_0 / L

Standing Wave A_1 (1,560 mm) & Node Cancellation

- **x_0 ≈ 1/9 L (173 mm):** Cancels sharp **H_9 (495 Hz)** at its node.
- **x_0 = 1/8 L (195 mm):** Node on **H_8 (−22 dB)**, tames dissonant H_7 & H_9.
- **x_0 = 1/7 L (223 mm):** Cancels natural 7th **H_7 (−22 dB)** for smooth warmth.

12 HARMONIC PARTIALS ON A_1 (55 HZ): MUSICAL INTERVAL & TIMBRAL ROLE

Partial	Interval	Timbral Role
H_1 (55 Hz)	Fundam.	Foundation & Projection
H_2 (110 Hz)	1st Oct	Warmth, Body & Volume
H_3 (165 Hz)	Twelfth	Sonorous Core & Richness
H_4 (220 Hz)	2nd Oct	Clear Octave Sheen
H_5 (275 Hz)	Maj 3rd	Cantabile & Singing Color
H_6 (330 Hz)	Perf 5th	Brilliance & Carry
H_7 (385 Hz)	Harm 7th	Dissonant 7th (rough)
H_8 (440 Hz)	3rd Oct	Presence & Radiance
H_9 (495 Hz)	Maj 2nd	Metallic Edge
H_{10} (550 Hz)	Maj 3rd	Bright Overtone Shimmer
H_{11} (605 Hz)	11th Part.	Sharp Inter-Harmonic
H_{12} (660 Hz)	Perf 5th	Crisp Attack Tip

The Hammerhead as a Complex Spring: From Cristofori to Zone Balance

How elasticity, 2:1 compaction (crown → ½ flank), and shoulder transition govern playable dynamics (pp–ff)

PHYSICAL MODELS & WORKSHOP DIAGNOSTIC STATES (PP ↔ FF)

✦ CRISTOFORI (1700)

Leathered Roll: Tension Creates Elasticity

Deer leather shapes gentle *piano*; ring tension (**88% bounce**, **τc ≈ 1.85 ms**) releases the string crisply.

Core Insight: Tension keeps τc short — dead stiffness lingers and damps the string.

✦ ZONE BALANCE (PP ↔ FF)

Supple Transition for pp — Elastic ½-Flank for ff

Force flows smoothly from **Crown (100%)** into the **½-Flank (50%)**: delicate *pp* and short **τc ≈ 1.80 ms** for full *ff* (**0.0 dB**).

Key Rule: Full dynamics emerge only when gentle onset and fast flank rebound work together.

▲ NO PIANISSIMO (PP)

Overly Hard Crown or Transition Blocks pp

Without initial cushion at the **Crown (125%)**, even light touch hits a hard shell (**+4.8 dB**, **τc ≈ 1.35 ms**) — no *pp*.

Remedy: Gently relieve the stiff transition behind the crown with 90° stitches.

▲ NO FORTISSIMO (FF)

Low Elasticity Prolongs Contact Time τc

Sluggish, inelastic felt (only **54% bounce**) lingers at **τc ≈ 2.95 ms** and damps out *ff*.

Physics: Carrying *ff* requires rapid elastic rebound, not rigid hardness.

▲ DOUBLE LOCK (NO PP & FF)

Hard Transition Kills pp and Decouples the Flank

Hard onset blocks *pp*; locking out the ½-flank (**18% active**) prolongs **τc to 2.75 ms** and chokes *ff*.

Diagnosis: Open the stiff outer contour to restore both *pp* cushion and flank rebound.

Voicing Zones & Needling Effect: The Renner Clock Model

Always perpendicular to the wool fiber (⊥): from the 9:00–3:00 wedges to the elasticity stitch above the rivet

ALL 6 RENNER CLOCK-MODEL VOICING ZONES & FIBER-COURSE RULE

✦ PRE-NEEDLING • BLUE FF-WEDGE (9–10 / 2–3H)

`7-9 mm • +3.2 dB`

Deep pre-needling releases factory rigidity and opens fortissimo

● **Blue ff-Wedge (9–10 / 2–3h) | 7–9 mm (pre-UF) · ⊥ fiber**

Pressing **perpendicular to the fiber course** up to the underfelt channels elasticity **upward into the f-wedge**.

> **Effect:** Boosts bass fullness (**H1–H4 +3.2 dB**) while keeping the crown intact.

✦ MID-VOICING • GREEN F-WEDGE (10–11 / 1–2H)

`4-6 mm • +2.4 dB`

Opens the shoulder zone for a singing cantabile core

● **Green f-Wedge (10–11 / 1–2h) | 4–6 mm · ⊥ fiber course**

Stitches **perpendicular to the fiber (4–6 mm)** relax stiff outer contours so the flank flexes freely for a **cantabile core**.

> **Effect:** Smooth *mf* → *f* growth and softer upper glare (**–2.5 dB**).

✦ BALANCE • ORANGE P-WEDGE (11–12 / 12–1H)

`2.5-4 mm • -3.8 dB`

Shallow shoulder needling shapes a warm, rounded piano tone

● **Orange p-Wedge (11–12 / 12–1h) | 2.5–4 mm · ⊥ fiber course**

2.5–4 mm stitches perpendicular to the fiber beside the crown cushion glassy *piano* partials while keeping *forte*.

> **Effect:** Evens tone color across 88 keys (**H8–H12 -3.8 dB**).

✦ FINE VOICING • PURPLE PP-WEDGE (≤ 2 MM)

`≤ 2.0 mm • -5.1 dB`

Micro-crown needling (≤ 2 mm) creates a velvety pp cushion

● **Purple pp-Wedge (Crown) | ≤ 2 mm · btw grooves**

Short single needle **max. 1–2 mm deep** perpendicular to the fiber and **strictly between string grooves**.

> **Rule:** Never prick in the grooves and never exceed **2 mm**!

✦ TENSION BUILD • ABOVE RIVET → FLANK

`10-14 mm • +3.1 dB`

Few, deep stitches boost elasticity and upper partials

● **Above Rivet → Flank | Few, deep stitches (10–14 mm) ↑**

Placing **few, deep stitches from just above the rivet toward the flank** tightens lower flank support (**τc ≈ 1.58 ms**).

> **Effect:** Increases **elasticity** and boosts **high partials (H7–H12 +3.1 dB)**.

▲ TABOO ZONE 12:00 • DEEP CROWN CORE

`× TABOO • -6.8 dB`

Deep crown stitches destroy surface tension irreversibly!

● **Red Taboo (12:00 Core) | ✕ TABOO: > 2 mm forbidden!**

Needling **> 2 mm into the red core triangle** severs the tension arch: rebound collapses and **τc jumps to 3.15 ms**.

> **Damage:** Tone turns **hollow and dead** — losing both *p* and *ff*!

▲ NEEDLING ERROR • AGAINST FIBER COURSE

Oblique stitches against the fiber course sever wool layers!

Working obliquely instead of **perpendicular to the fiber** cuts tensioned bridges, breaking force flow and **prolonging τc**.

> **Core Law:** Always follow the **fiber course** and press **perpendicular to the fiber**!

Crown & String Plane in 3-String Choirs: Phase & Cancellation

Even a 0.2 mm height or angle mismatch causes phase cancellation notches across partials H_1–H_{12}

ALL 4 UNISON & CROWN GEOMETRY STATES (3-STRING CHOIR)

✦ TARGET STATE • FLAT & PARALLEL

All 3 Unison Strings Launch Exactly in Phase ($\Delta t = 0$ ms, $\Delta\varphi = 0°$)

When a cleanly filed crown meets a level 3-string unison squarely, all three string waves add constructively to 3× total amplitude.

> **Tonal Effect:** Carrying fundamental core (H_1–H_4) and a radiant harmonic spectrum with zero comb-filter notches.

✕ FAULT 1 • TILTED CROWN

Staggered Contact ($S_1 \rightarrow S_2 \rightarrow S_3$) Cancels Upper Partials

A tilted crown strikes $S_1 \rightarrow S_2 \rightarrow S_3$ sequentially. On upper partials, $\Delta\varphi$ reaches 120°–240°: wave crests cancel wave troughs!

> **Effect & Remedy:** Metallic buzz & spectral notches. Square hammer travel and file the crown parallel first!

✕ FAULT 2 • UNLEVELED STRING PLANE

One Offset String Fights the Unison at 180° Out of Phase

If one string sits just 0.2–0.3 mm high or low at the agraffe or bridge, it launches late and subtracts energy at $\Delta\varphi \approx 180°$.

> **Effect & Remedy:** Hollow tone & false beats despite pure tuning. Level strings at agraffes and seat bridge pins first!

✕ FAULT 3 • HOLLOW CROWN / WORN GROOVES

Outer Strings Lead While Center String Lags Out of Phase

Outer strings S_1 and S_3 speak early while center string S_2 lags weakly—dropping out in pp and canceling out of phase in forte.

> **Golden Rule:** No needling can fix phase cancellation—file and shape the crown (320–400 grit) first!

String Preparation: Hitch Pins, Bridge Pins & Agraffes

“Even the finest hammerhead can only bring out what is prepared in the instrument!”

STRING BEDDING

`4 Prep Steps`

Exact String Seating Before the First Needle Stitch

Only firmly seated, level-aligned, and purely tuned string unisons allow an honest tonal diagnosis of the hammerhead.

- **1. Hitch Pins & Bearing:** Clean bearing surfaces and check firm, tension-free string seating at the hitch pins.
- **2. Seat at Bridge Pins:** Gently tap strings onto the bridge using a brass drift punch and light hammer to eliminate false beats.
- **3. Agraffes & String Plane:** Level and space the 3 unison strings cleanly behind the agraffes using fine pliers.
- **4. Unison & Damper Check:** Verify simultaneous hammer contact, clean damper seating, and 100% beatless unison tuning.

> ✦ Golden Rule: No needle can fix an unevenly seated or unlevel string unison!

VISUAL & WORKSHOP REFERENCE CAPTIONS

Exact String Seating Before the First Needle Stitch

- **01 • HITCH PINS:** Cleaning bearing surfaces & checking seating at hitch pins
- **02 • BRIDGE PINS:** Seating strings cleanly at bridge pins with brass punch
- **03 • AGRAFFES:** Spacing and leveling the 3-string choir behind the agraffes
- **04 • UNISON CHECK:** Checking damper seating, pure unison & simultaneous contact

Intermediate, Fine & Una Corda Voicing

Chromatic equalization, max. 2 mm crown needling & una corda shift-pedal voicing

FINE & SHIFT VOICING

`Max. 2 mm • Una Corda`

Concert-Grade Equalization—Including the Shift Pedal

All 88 notes blend chromatically—both open and with shift pedal depressed—into a seamless, noise-free tonal progression.

- **1. BDK Fine Holder:** Short single needle in the ergonomic BDK holder (43-0003269) for tactile crown control.
- **2. Mid-Voicing (Shoulder Zone):** Targeted evening-out of outliers in the forte and piano wedges on a firm support pad.
- **3. 2 mm Crown Needling:** With Chop-Stick (43-0003271) between strings – max. 2 mm deep strictly between string grooves!
- **4. Smooth the Crown:** Dress raised wool fibers cleanly with a fine sanding paddle (320–400 grit) for clear attack.
- **5. Shift-Pedal Voicing (Una Corda):** With pedal depressed, voice only contacted string zones & lightly chamfer the left crown edge.

> ✦ Micro-dosage at the crown & left edge ensures silky shimmer even in una corda!

VISUAL & WORKSHOP REFERENCE CAPTIONS

Concert-Grade Equalization—Including the Shift Pedal

- **01 • BDK HOLDER:** Short single needle in the BDK holder for delicate crown control
- **02 • SHOULDER ZONE:** Targeted needling in the piano and forte wedges on support pad
- **03 • MAX. 2 MM CROWN NEEDLING:** Chop-Stick in the grand: max. 2 mm crown needling strictly between grooves
- **04 • DRESSING & UNA CORDA:** Fine crown dressing (320–400 grit) & light left-edge chamfer for una corda

Lacquer Dosing, Heat Ironing (max. 135 °C), Crown Finish & Weight

Interactive zone effects of lacquer hardeners, thermal smoothing, micro-polishing (≤ 15 µm) & register-specific hammer mass

01 • LACQUER DOSING ZONES & TIME-COMPACTION BEHAVIOR

✦ FOUNDATION SUPPORT • RIVET AREA

H_1–H_4 +3.4 dB

Lacquer at the rivet reinforces the foundation and can lift felt tension

Applied in the rivet area, lacquer stiffens the lower hammer body and—depending on lacquer type—lifts felt tension while leaving flank elasticity untouched.

> ✓ **Advantage:** Located below the hammer moulding — the elastic impact arch stays protected.

✦ ABOVE HAMMER MOULDING • LATERAL APPLICATION

H_9–H_{12} +3.8 dB

Lateral dosing above the moulding boosts high partials — and compacts over time!

Lightly applied from the sides above the hammer moulding, diluted lacquer reinforces the upper bridge to boost high partials (H_7–H_{12}) while keeping the outer flank free to flex.

> ▸ **Time Factor:** Every application above the hammer moulding compacts further over time during play — always leave a safety margin!

▲ OVERDOSE ERROR • EXCESS LATERAL LACQUER

H_1–H_4 -3.8 dB

Too much lacquer glues the connection to the flank and disrupts spring action!

When over-applied above the hammer moulding, lacquer wicks outward into the flank and glues wool layers together — locking up the hammerhead's complex spring mechanism.

> ✕ **Result:** Disrupted spring function, lost dynamic range, and progressive hardening over time!

✕ STRICTLY AVOID • LACQUER DIRECTLY ON CROWN

Nasal +6.0 dB

Lacquer on the crown creates glassy, nasal hardness and permanent damage!

Applying hardener directly to the 12:00 strike crown forms a brittle crust in the string contact zone, destroying surface elasticity while progressive strike compaction worsens the glare.

> ✕ **Workshop Rule:** Avoid lacquer on the crown — risk of permanent hammerhead damage!

✦ BOOSTING FUNDAMENTAL TONE & PROJECTION

Control via register-specific hammerhead weight (instead of over-hardening):

- **Middle to Bass → Increase Weight (+m):** Higher hammerhead mass drives greater energy into heavy strings, boosting fundamental fullness and punch.
- **Toward Treble → Reduce Weight (−m):** Lower mass accelerates faster and shortens contact time on short treble strings for clear, carrying projection.

02 • HEAT IRONING

`Strictly max. 135 °C`

Heat Ironing the Heads (up to max. 135 °C)

Smoothing with a heated iron lays down raised fibers, compacts the outer zone, and lifts shell tension.

- **Max. 135 °C Limit:** Never iron hotter – higher heat damages wool keratin and natural lanolin.
- **Smooth Motion:** Glide evenly along the contour so internal spring elasticity stays intact.

> ✦ Thermostat required: Max. 135 °C prevents fiber damage & scorching

03 • ULTRA-FINE CROWN

`≤ 15 µm • Braun Series 1`

Micro-Polished Crown (≤ 15 µm) & Shaving

An ultra-smooth crown removes damping fuzz fibers and optimizes string-contact hardness without chemicals.

- **Micro-Abrasive down to 15 µm (15 my):** Polishes the crown mirror-smooth for crisp string contact.
- **Braun Series 1 Foil Shaver:** Standard foil shaver trims protruding fibers with zero felt loss.

> ✦ Gentlest optimization: Pure tone & 100% preserved spring function

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